BUSINESS REQUIREMENTS DOCUMENT

Automated Attendance Management System

College Portal | MIT World Peace University

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1. Executive Summary

MIT World Peace University (MIT-WPU) currently manages student attendance through a fragmented, largely manual process. Faculty mark attendance on paper registers or update records individually in the college portal, leading to significant delays, data entry errors, and lack of real-time visibility for students and administrators.

This Business Requirements Document (BRD) defines the functional and non-functional requirements for an Automated Attendance Management System (AAMS) to be integrated into the existing college portal. The proposed system will enable digital attendance marking by faculty, provide students with real-time subject-wise attendance dashboards, and trigger automated alerts when attendance falls below the 75% university threshold.

Expected outcomes include elimination of manual attendance errors, a 40% reduction in student eligibility disputes, improved faculty productivity, and a significant decrease in administrative workload for the examination and academic affairs departments.

Business Problem Statement

The current attendance management process at MIT-WPU suffers from the following critical issues:

- Attendance is marked manually on paper or entered individually into the portal, causing delays of up to 48 hours
- Students have no real-time visibility into their subject-wise attendance percentage
- No automated alert is triggered when attendance drops below the 75% eligibility threshold
- Faculty have no consolidated dashboard to track class-level attendance trends
- The examination department manually compiles attendance data to determine exam eligibility, a process prone to errors
- No audit trail exists for disputed attendance records, leading to lengthy resolution processes

2. Project Scope

2.1 In Scope

- Digital attendance marking by faculty via web portal
- Real-time subject-wise attendance dashboard for students
- Automated email and SMS alerts when student attendance falls below 80% (warning) and 75% (critical)
- Faculty dashboard showing class-wise and student-wise attendance trends
- Admin and examination department dashboard for bulk attendance reports
- Attendance correction and dispute request workflow with full audit trail
- Exportable attendance reports in PDF and Excel formats
- Integration with the existing college portal login and student database

2.2 Out of Scope

- Biometric or RFID-based automatic attendance hardware integration
- Changes to the existing fee payment or results portal
- Timetable and scheduling module development
- Mobile app for Android and iOS (deferred to Phase 2)

2.3 Assumptions

- All faculty members have access to a computer or smartphone with internet connectivity during or after class
- The existing college portal has an active student and faculty database accessible via API
- Students have registered mobile numbers and email addresses in the college system
- The university minimum attendance threshold of 75% remains unchanged

2.4 Constraints

- System must comply with the Information Technology Act 2000 and applicable Indian data privacy regulations
- Solution must work on existing devices with no new hardware procurement in Phase 1
- System must be live before commencement of the next academic semester
- Portal must support a minimum of 5,000 concurrent users during peak hours (9 AM to 5 PM)

- System must function on low-bandwidth connections of 2 Mbps or lower

3. Stakeholder Analysis

The following stakeholders have been identified for the Automated Attendance Management System project:

Stakeholder	Role	Key Interests / Expectations	Impact
Student	Primary End User	View real-time subject-wise attendance; receive early alerts before falling below 75%	High
Faculty Member	Data Entry & Oversight	Mark attendance quickly; view class-level attendance trends without manual effort	High
Class Coordinator	Academic Operations	Monitor overall class attendance; flag at-risk students; generate periodic reports	High
Examination Department	Compliance & Eligibility	Generate accurate eligibility reports; eliminate manual compilation errors before exams	High
Academic Affairs / Admin	Policy & Governance	Institutional reporting, accreditation data, compliance with UGC norms	High
IT Department	Technical Ownership	Clean API integration, system performance, security, and maintainability	Medium
Parents / Guardians	Secondary Stakeholder	Optional alerts when student attendance is critically low	Low
University Management	Executive Sponsor	Improved institutional metrics, reduced disputes, accreditation readiness	Medium

4. Functional Requirements

Functional requirements define the specific behaviours and features the system must provide. Each requirement is assigned a unique ID and priority (Must Have / Should Have / Nice to Have).

4.1 Student-Facing Requirements

Req ID	Requirement Description	Priority
FR-01	The system shall display a real-time subject-wise attendance dashboard showing attendance percentage for each enrolled subject.	Must Have
FR-02	The system shall show the number of classes held, attended, and remaining for each subject in the current semester.	Must Have
FR-03	The system shall send an automated email and SMS early warning alert when attendance in any subject falls below 80%.	Must Have

FR-04	The system shall send a critical alert when attendance in any subject falls below the 75% eligibility threshold.	Must Have
FR-05	The system shall display a subject-wise attendance history log showing date, lecture number, and present or absent status.	Must Have
FR-06	The system shall allow a student to raise an attendance correction request for a specific date and subject with supporting remarks.	Should Have
FR-07	The system shall display a projection showing how many future classes must be attended to reach or maintain 75%.	Should Have
FR-08	The system shall allow students to download their attendance report as a PDF for a selected date range.	Nice to Have

4.2 Faculty-Facing Requirements

Req ID	Requirement Description	Priority
FR-09	The system shall allow faculty to mark attendance by selecting subject, date, and lecture slot, then marking each student as Present, Absent, or Late.	Must Have
FR-10	The system shall allow faculty to mark all students present in bulk, then individually unmark absentees, to reduce data entry time.	Must Have
FR-11	The system shall allow faculty to edit attendance within 24 hours of submission, with all changes logged in the audit trail.	Must Have
FR-12	The system shall display a class-level dashboard showing overall attendance per subject and flagging students below 75%.	Must Have
FR-13	The system shall allow faculty to approve or reject student correction requests with mandatory remarks.	Should Have
FR-14	The system shall allow faculty to export subject-wise student attendance reports in Excel format.	Should Have

4.3 Admin / Examination Department Requirements

Req ID	Requirement Description	Priority
FR-15	The system shall generate a semester-end eligibility report listing students eligible and ineligible for examinations per subject.	Must Have
FR-16	The system shall allow administrators to set and update the minimum attendance threshold at department or university level.	Must Have
FR-17	The system shall provide a consolidated attendance dashboard across all departments, batches, and subjects.	Must Have
FR-18	The system shall maintain a complete audit log of all entries, edits, and corrections with timestamps and user IDs.	Must Have
FR-19	The system shall support bulk import of timetable and enrollment data via CSV to initialise a new semester.	Should Have
FR-20	The system shall allow admins to generate reports filtered by department, batch, subject, or date range.	Should Have

5. Non-Functional Requirements

Req ID	Category	Requirement	Target / Metric
NFR-01	Performance	Attendance dashboard page load time for all users	Under 3 seconds
NFR-02	Performance	Attendance submission confirmation response time	Under 2 seconds
NFR-03	Scalability	Concurrent users during peak academic hours (9 AM to 5 PM)	Min 5,000 users
NFR-04	Availability	System uptime during academic working days	99.5% SLA
NFR-05	Security	All data transmitted between client and server	TLS 1.2 or higher
NFR-06	Security	Role-based access: students see only their data; faculty see only their subjects	Enforced at API level
NFR-07	Data Integrity	Attendance records locked after 24-hour window; not modifiable without admin approval	Immutable after lock
NFR-08	Usability	First-time faculty user completes attendance marking without training	Under 3 minutes
NFR-09	Compatibility	Portal fully functional on Chrome, Firefox, and Edge	Latest 2 browser versions
NFR-10	Accessibility	System usable on screens with resolution as low as 1024x768	Responsive layout required
NFR-11	Reliability	Offline attendance marking syncs correctly when connectivity is restored	Offline mode required
NFR-12	Compliance	All student data stored and processed per India IT Act 2000	Legal review required

6. User Stories

User stories follow the standard Agile format: As a [user], I want to [action], so that [benefit]. Each story includes acceptance criteria defining when the story is complete.

6.1 Student User Stories

ID	User Story	Priority	Acceptance Criteria
US-01	As a student, I want to see my subject-wise attendance percentage in real time so that I can track my eligibility before exams.	High	Dashboard shows % per subject; updates within 60 min of faculty submission

US-02	As a student, I want an SMS and email alert when my attendance drops below 80% so that I can take corrective action early.	High	Alert sent within 30 min of threshold breach; includes subject name and current %
US-03	As a student, I want to see a class-by-class attendance history so that I can verify which classes I was marked absent for.	High	History shows date, subject, slot, and P or A status for the full semester
US-04	As a student, I want to raise a correction request for a wrongly marked absence so that disputes can be resolved without visiting the department.	Medium	Request form accepts date, subject, remarks; confirmation shown on submission
US-05	As a student, I want to see how many classes I must attend to reach 75% so that I can plan my attendance going forward.	Medium	Projection shows classes needed; updates dynamically with each new submission

6.2 Faculty User Stories

ID	User Story	Priority	Acceptance Criteria
US-06	As a faculty member, I want to mark attendance for my class in under 2 minutes so that it does not disrupt the lecture.	High	Bulk mark-all-present available; faculty unmarks absentees in one step
US-07	As a faculty member, I want to edit an attendance entry within 24 hours so that I can correct mistakes without admin intervention.	High	Edit option active for 24 hours post-submission; all changes logged
US-08	As a faculty member, I want to see which students are below 75% so that I can counsel them proactively.	High	Dashboard highlights at-risk students in red; sortable by attendance %
US-09	As a faculty member, I want to approve or reject correction requests with remarks so that the process is transparent and traceable.	Medium	Approve or Reject buttons in faculty portal; remark field is mandatory

6.3 Admin / Examination Department User Stories

ID	User Story	Priority	Acceptance Criteria
US-10	As an exam coordinator, I want to generate a subject-wise eligibility report at semester end so that I can finalise the exam hall ticket list accurately.	High	Report lists eligible and ineligible students per subject; exports to Excel and PDF

US-11	As an admin, I want to view attendance across all departments in one dashboard so that I can monitor institutional compliance.	High	Consolidated view with filters for department, batch, subject, and date range
US-12	As an admin, I want a full audit trail of all entries and corrections so that I can resolve disputes with documented evidence.	High	Audit log shows user ID, action, timestamp, and before or after values for every change

7. Process Flow Diagrams

The AS-IS and TO-BE flows document how attendance management currently works versus the proposed automated state. Visual draw.io diagrams should be created separately using these flows as reference.

7.1 AS-IS Process (Current State)

How attendance is currently managed and where it fails

Step	Actor	Action	Pain Point
1	Faculty	Takes attendance verbally or on paper register during class	No digital record created at time of class
2	Faculty	Manually enters attendance into college portal hours or days later	Delays up to 48 hours; prone to data entry errors
3	Student	Logs into portal to check attendance	Data may be outdated or missing; no alerts sent
4	Student	Realises attendance is below 75% only when filling exam form	Too late to take corrective action; causes eligibility disputes
5	Student	Visits department office to raise attendance dispute	Manual, time-consuming; no audit trail exists
6	Faculty / Admin	Manually verifies register and updates portal if correction approved	No formal workflow; corrections untraceable
7	Exam Dept.	Manually compiles attendance before each exam to generate eligibility list	Error-prone; time-intensive; delays exam form issuance

7.2 TO-BE Process (Proposed State)

How the automated system will work end-to-end

Step	Actor	Action	Improvement
1	Faculty	Opens portal during or after class; selects subject and lecture slot	Digital process; no paper register required

2	Faculty	Uses bulk mark-present; unmarks absentees; submits in under 2 minutes	Attendance recorded in real time; errors minimised
3	System	Instantly updates student dashboard with new percentage per subject	Students see updated data within 60 minutes of submission
4	System	Detects attendance below 80%; triggers early warning email and SMS	Student warned early enough to attend more classes and recover
5	System	Detects attendance below 75%; triggers critical eligibility alert	No surprises at exam time; student and parent notified
6	Student	Raises online correction request with remarks for any incorrect absence	Formal digital workflow with audit trail; no office visit needed
7	Faculty	Reviews and approves or rejects correction request with mandatory remarks	Transparent process; all actions logged with timestamps
8	Exam Dept.	Generates one-click eligibility report filtered by subject, batch, or department	Zero manual compilation; accurate and instant report generation

8. Use Cases

Detailed use cases are documented for the three most critical interactions in the system.

Use Case 1 - Faculty Marks Class Attendance

Field	Detail
Use Case ID	UC-01
Use Case Name	Mark Class Attendance
Actor	Faculty Member
Precondition	Faculty is logged into the portal and has at least one active subject assigned for the current semester
Trigger	Faculty selects Mark Attendance from the portal navigation during or after a class
Main Flow	1. Faculty selects subject, date, and lecture slot 2. System loads full student list for that subject 3. Faculty clicks Mark All Present to pre-fill all as present 4. Faculty unchecks absent students 5. Faculty clicks Submit 6. System saves record and shows success confirmation 7. Student dashboards update within 60 minutes
Alternate Flow	If no internet: App saves attendance locally and syncs when connectivity is restored
Postcondition	Attendance recorded; dashboards updated; alerts triggered if any student breaches threshold
Business Value	Eliminates paper registers; reduces faculty admin time by over 60%

Use Case 2 - Student Views Attendance Dashboard and Receives Alert

Field	Detail
Use Case ID	UC-02
Use Case Name	View Attendance Dashboard and Receive Threshold Alert
Actor	Student
Precondition	Student is enrolled in the current semester and faculty has submitted at least one attendance record
Trigger	Student logs into the college portal and navigates to the Attendance module
Main Flow	1. Student opens Attendance module 2. System displays a card per subject: attendance %, classes held, classes attended 3. Student clicks a subject card to view class-by-class history 4. If subject is below 80%, yellow early warning banner is shown 5. If subject is below 75%, red critical banner shown and email plus SMS triggered 6. Student views projection of classes needed to recover to 75%
Alternate Flow	If no attendance submitted yet: System shows No attendance data available for this semester
Postcondition	Student is fully informed of their status and can take corrective action proactively
Business Value	Reduces exam eligibility disputes by 40%; improves student accountability

Use Case 3 - Student Raises Attendance Correction Request

Field	Detail
Use Case ID	UC-03
Use Case Name	Attendance Correction Request Workflow
Actor	Student (initiator), Faculty (approver)
Precondition	Student has been marked absent for a class they attended; record is visible in attendance history
Trigger	Student clicks Raise Correction Request next to the disputed entry in their history
Main Flow	1. Student selects date, subject, and lecture slot for the correction 2. Student fills in remarks explaining the dispute 3. Student submits the request 4. System notifies the faculty member via portal and email 5. Faculty reviews and clicks Approve or Reject with mandatory remarks 6. System updates the record if approved and notifies the student 7. All actions logged in audit trail with timestamps
Alternate Flow	If faculty does not respond within 5 working days: Request escalated to Class Coordinator automatically
Postcondition	Dispute resolved with full audit trail; no physical office visit required; record updated or rejected with documented reason
Business Value	Eliminates manual dispute handling; creates transparent, auditable correction process

9. Acceptance Criteria

The following criteria must be satisfied before the Automated Attendance Management System is considered complete and ready for production deployment.

9.1 Functional Acceptance Criteria

AC ID	Feature	Acceptance Criteria	Pass / Fail
AC-01	Attendance Marking	Faculty marks attendance for 60 students in under 2 minutes using bulk mark feature	—
AC-02	Real-Time Dashboard	Student dashboard reflects latest submission within 60 minutes of faculty submitting	—
AC-03	Early Warning Alert	Email and SMS sent within 30 minutes when attendance falls below 80%	—
AC-04	Critical Alert	Critical email, SMS, and in-portal banner triggered when attendance falls below 75%	—
AC-05	Attendance History	Student views class-by-class history with date, subject, slot, and P or A status for full semester	—
AC-06	Correction Workflow	Correction request reaches faculty within 15 minutes of student submission	—
AC-07	Correction Approval	Approved correction updates attendance record within 30 minutes; logged in audit trail	—
AC-08	Eligibility Report	Exam department generates subject-wise eligibility report in under 30 seconds; exports to Excel and PDF	—
AC-09	Audit Trail	Every entry, edit, and correction logged with user ID, action type, timestamp, and before or after values	—
AC-10	Edit Window Lock	Faculty can edit within 24 hours; edit option disabled and record locked after 24-hour window closes	—

9.2 Non-Functional Acceptance Criteria

AC ID	Category	Acceptance Criteria	Pass / Fail
AC-11	Performance	Dashboard loads within 3 seconds for all users under 5,000 concurrent user load test	—
AC-12	Availability	System achieves 99.5% uptime over 30-day monitoring period during academic working hours	—
AC-13	Security	Role-based access verified: students cannot see other students data; confirmed via security testing	—
AC-14	Usability	First-time faculty user completes attendance marking without training within 3 minutes	—
AC-15	Offline Mode	Attendance marked offline syncs correctly within 5 minutes of internet connectivity restored	—

10. Assumptions and Constraints

10.1 Assumptions

ID	Assumption
A-01	All faculty members have access to a smartphone or computer with internet connectivity during or immediately after class.
A-02	The existing college portal has an active and accurate student and faculty database accessible via API integration.
A-03	Students and faculty have valid and registered email addresses and mobile numbers in the college system.
A-04	The minimum attendance threshold of 75% is a fixed university policy and will not change during the project scope.
A-05	The IT department will provide API documentation for the existing student information system within the first two weeks of the project.
A-06	Faculty are willing to adopt the digital attendance process after a brief orientation session.

10.2 Constraints

ID	Constraint
C-01	The system must comply with the Information Technology Act 2000 and applicable Indian data privacy regulations.
C-02	No new hardware will be procured for Phase 1; the solution must work entirely on existing faculty and student devices.
C-03	The system must be live and operational before commencement of the next academic semester.
C-04	The portal must support a minimum of 5,000 concurrent users without performance degradation during peak hours.
C-05	The system must function correctly on low-bandwidth connections of 2 Mbps or lower to support campus network constraints.

11. Glossary

Term	Definition
BRD	Business Requirements Document - a formal document describing the business solution for a project.
AAMS	Automated Attendance Management System - the proposed system defined in this BRD.

75% Threshold	The minimum attendance percentage mandated by MIT-WPU for a student to be eligible to appear in semester examinations.
AS-IS Process	The current state of a business process before any improvements are implemented.
TO-BE Process	The desired future state of a business process after the proposed system is deployed.
Functional Req.	A requirement specifying what the system must do - its features and behaviours.
Non-Functional Req.	A requirement specifying how well the system performs - covering quality attributes like speed, security, and reliability.
User Story	A short description of a feature from an end user perspective: As a [user], I want [action], so that [benefit].
Acceptance Criteria	Conditions that must be met for a feature to be considered complete and ready for sign-off.
Audit Trail	A chronological record of all system activities including who performed an action, what was changed, and when.
Role-Based Access	A security model restricting system access based on a user role: student, faculty, or admin.
Bulk Mark	A feature allowing faculty to mark all students present in one click, then individually unmark absentees.
Correction Request	A formal workflow allowing a student to dispute an incorrect absence marking with faculty review and approval.
Eligibility Report	A report listing students who meet or fail to meet the attendance threshold for examination entry.
TLS	Transport Layer Security - a cryptographic protocol providing secure encrypted communication over the internet.

12. Document Sign-Off

This BRD requires review and formal approval from the following stakeholders before development work commences.

Role	Name	Signature	Date
Business Analyst	Akshat Agrawal	_____	_____
Academic Affairs Head	_____	_____	_____
IT Department Lead	_____	_____	_____
Examination Controller	_____	_____	_____
Faculty Representative	_____	_____	_____

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